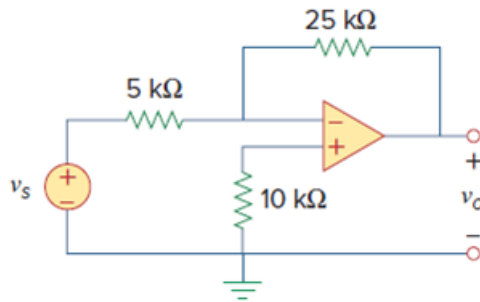


Problem

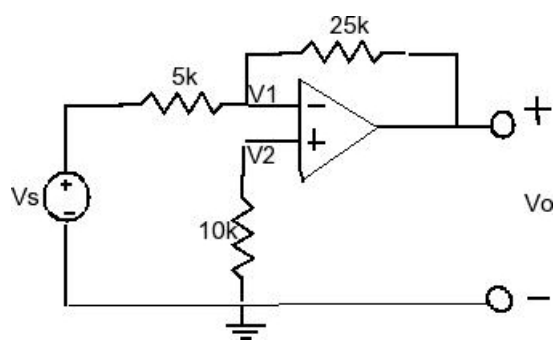
Calculate the voltage ratio v_o/v_s for the op amp circuit of Fig. 5.51. Assume that the op amp is ideal.

Figure. 5.51:



Step-by-step solution

Step 1 of 1



Using KCL at node V1 and V2 will allow V_o / V_s ratio

KCL at V2

$$\frac{V_2 - 0}{10} = 0$$

$$V_2 = 0$$

Since the op amp is ideal $V_1 = V_2$

$$V_1 = 0$$

KCL at V1

$$\frac{V_1 - V_s}{5} + \frac{V_1 - V_o}{25} = 0$$

remember $V_1 = 0$

simplifying we get

$$-5V_s - V_o = 0$$

$$V_o = -5V_s$$

$$V_o / V_s = -5$$